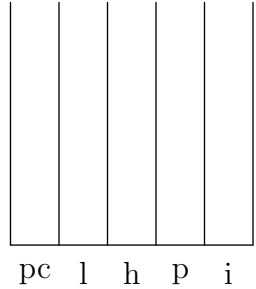
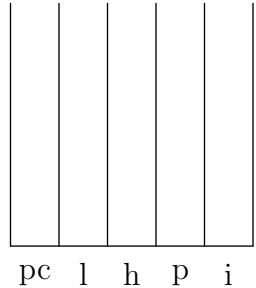
```
quickSort(a, l, h) {
  if(h-l>3) {
    p = choosePivot(a, l, h)
    i = partition(a, p, l, h)
    quickSort(a, l, i-1)
    quickSort(a, i+1, h)
  } else
    insertionSort(a, l, h)
  return
}
```



```
0 quickSort(a, 0, 19) {
1   if(19-0>3) {
2     p = choosePivot(a, 0, 19)
3     i = partition(a, p, 0, 19)
4     quickSort(a, 0, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 0, 19)
8   return
9 }
```

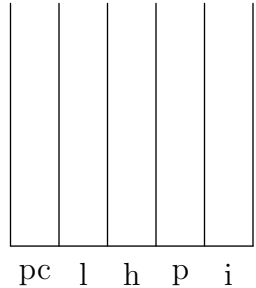
```
PC = 0
l  = 0
h  = 19
p  = #
i  = #
```



0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
23	48	40	41	46	32	24	50	73	57	18	58	69	45	37	3	95	57	81	28
low																			high

```
0 quickSort(a, 0, 19) {
1   if(19-0>3) {
2     p = choosePivot(a, 0, 19)
3     i = partition(a, p, 0, 19)
4     quickSort(a, 0, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 0, 19)
8   return
9 }
```

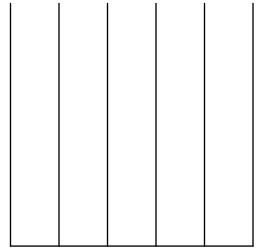
```
PC = 1
l  = 0
h  = 19
p  = #
i  = #
```



0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
23	48	40	41	46	32	24	50	73	57	18	58	69	45	37	3	95	57	81	28
low																			high

```
0 quickSort(a, 0, 19) {
1   if(19-0>3) {
2     p = choosePivot(a, 0, 19)
3     i = partition(a, p, 0, 19)
4     quickSort(a, 0, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 0, 19)
8   return
9 }
```

```
PC = 2
l  = 0
h  = 19
p  = #
i  = #
```



0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
23	48	40	41	46	32	24	50	73	57	18	58	69	45	37	3	95	57	81	28
low																			high

```
0 quickSort(a, 0, 19) {
1   if(19-0>3) {
2     p = choosePivot(a, 0, 19)
3     i = partition(a, p, 0, 19)
4     quickSort(a, 0, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 0, 19)
8   return
9 }
```

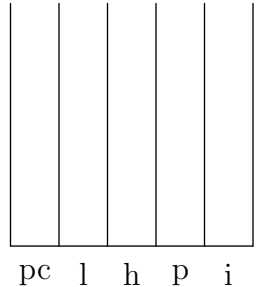
```
PC = 2
l  = 0
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
2	0	19	#	#

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
23	48	40	41	46	32	24	50	73	57	18	58	69	45	37	3	95	57	81	28
low																			high

```
0 quickSort(a, 0, 19) {
1   if(19-0>3) {
2     p = choosePivot(a, 0, 19)
3     i = partition(a, 28, 0, 19)
4     quickSort(a, 0, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 0, 19)
8   return
9 }
```

```
PC = 3
l  = 0
h  = 19
p  = 28
i  = #
```



0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
23	48	40	41	46	32	24	50	73	57	18	58	69	45	37	3	95	57	81	28
low																			high

```
0 quickSort(a, 0, 19) {
1   if(19-0>3) {
2     p = choosePivot(a, 0, 19)
3     i = partition(a, 28, 0, 19)
4     quickSort(a, 0, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 0, 19)
8   return
9 }
```

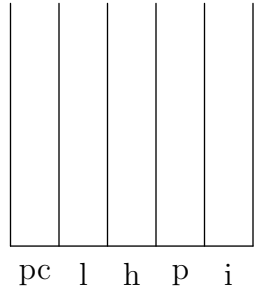
```
PC = 3
l  = 0
h  = 19
p  = 28
i  = #
```

pc	l	h	p	i
3	0	19	28	#

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
23	3	18	24	**28**	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
low																			high

```
quickSort(a, 0, 19) {
  if(19-0>3) {
    p = choosePivot(a, 0, 19)
    i = partition(a, 28, 0, 19)
    quickSort(a, 0, 4-1)
    quickSort(a, 4+1, 19)
  } else
    insertionSort(a, 0, 19)
  return
}
```

```
PC = 4
l  = 0
h  = 19
p  = 28
i  = 4
```



0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
23	3	18	24	28	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
low																			high

```
0 quickSort(a, 0, 3) {
1   if(3-0>3) {
2     p = choosePivot(a, 0, 3)
3     i = partition(a, p, 0, 3)
4     quickSort(a, 0, i-1)
5     quickSort(a, i+1, 3)
6   } else
7     insertionSort(a, 0, 3)
8   return
9 }
```

```
PC = 0
l  = 0
h  = 3
p  = #
i  = #
```

pc	l	h	p	i
4	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
23	3	18	24	28	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
low			high																

```
0 quickSort(a, 0, 3) {
1   if(3-0>3) {
2     p = choosePivot(a, 0, 3)
3     i = partition(a, p, 0, 3)
4     quickSort(a, 0, i-1)
5     quickSort(a, i+1, 3)
6   } else
7     insertionSort(a, 0, 3)
8   return
9 }
```

```
PC = 1
l  = 0
h  = 3
p  = #
i  = #
```

pc	l	h	p	i
4	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
23	3	18	24	28	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
low			high																

```
0 quickSort(a, 0, 3) {
1   if(3-0>3) {
2     p = choosePivot(a, 0, 3)
3     i = partition(a, p, 0, 3)
4     quickSort(a, 0, i-1)
5     quickSort(a, i+1, 3)
6   } else
7     insertionSort(a, 0, 3)
8   return
9 }
```

```
PC = 7
l  = 0
h  = 3
p  = #
i  = #
```

pc	l	h	p	i
4	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
23	3	18	24	28	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
low			high																

```
0 quickSort(a, 0, 3) {
1   if(3-0>3) {
2     p = choosePivot(a, 0, 3)
3     i = partition(a, p, 0, 3)
4     quickSort(a, 0, i-1)
5     quickSort(a, i+1, 3)
6   } else
7     insertionSort(a, 0, 3)
8   return
9 }
```

```
PC = 7
l  = 0
h  = 3
p  = #
i  = #
```

pc	l	h	p	i
7	0	3	#	#
4	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
low			high																

```
0 quickSort(a, 0, 3) {
1   if(3-0>3) {
2     p = choosePivot(a, 0, 3)
3     i = partition(a, p, 0, 3)
4     quickSort(a, 0, i-1)
5     quickSort(a, i+1, 3)
6   } else
7     insertionSort(a, 0, 3)
8   return
9 }
```

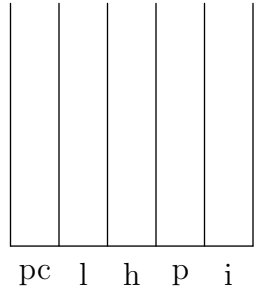
```
PC = 8
l  = 0
h  = 3
p  = #
i  = #
```

pc	l	h	p	i
4	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
low			high																

```
0 quickSort(a, 0, 19) {
1   if(19-0>3) {
2     p = choosePivot(a, 0, 19)
3     i = partition(a, 28, 0, 19)
4     quickSort(a, 0, 4-1)
5     quickSort(a, 4+1, 19)
6   } else
7     insertionSort(a, 0, 19)
8   return
9 }
```

```
PC = 5
l  = 0
h  = 19
p  = 28
i  = 4
```



0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
low																			high

```
quickSort(a, 5, 19) {
  if(19-5>3) {
    p = choosePivot(a, 5, 19)
    i = partition(a, p, 5, 19)
    quickSort(a, 5, i-1)
    quickSort(a, i+1, 19)
  } else
    insertionSort(a, 5, 19)
  return
}
```

```
PC = 0
l  = 5
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
					low														high

```
0 quickSort(a, 5, 19) {
1   if(19-5>3) {
2     p = choosePivot(a, 5, 19)
3     i = partition(a, p, 5, 19)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 5, 19)
8   return
9 }
```

```
PC = 1
l  = 5
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
					low														high

```
0 quickSort(a, 5, 19) {
1   if(19-5>3) {
2     p = choosePivot(a, 5, 19)
3     i = partition(a, p, 5, 19)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 5, 19)
8   return
9 }
```

```
PC = 2
l  = 5
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
					low														high

```
0 quickSort(a, 5, 19) {
1   if(19-5>3) {
2     p = choosePivot(a, 5, 19)
3     i = partition(a, p, 5, 19)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 5, 19)
8   return
9 }
```

```
PC = 2
l  = 5
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
2	5	19	#	#
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
					low														high

```
0 quickSort(a, 5, 19) {
1   if(19-5>3) {
2     p = choosePivot(a, 5, 19)
3     i = partition(a, 46, 5, 19)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 5, 19)
8   return
9 }
```

```
PC = 3
l  = 5
h  = 19
p  = 46
i  = #
```

pc	l	h	p	i
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	50	73	57	40	58	69	45	37	48	95	57	81	46
					low														high

```
0 quickSort(a, 5, 19) {
1   if(19-5>3) {
2     p = choosePivot(a, 5, 19)
3     i = partition(a, 46, 5, 19)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 5, 19)
8   return
9 }
```

```
PC = 3
l  = 5
h  = 19
p  = 46
i  = #
```

pc	l	h	p	i
3	5	19	46	#
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	37	45	40	46	58	69	73	50	48	95	57	81	57
					low														high

```
0 quickSort(a, 5, 19) {
1   if(19-5>3) {
2     p = choosePivot(a, 5, 19)
3     i = partition(a, 46, 5, 19)
4     quickSort(a, 5, 10-1)
5     quickSort(a, 10+1, 19)
6   } else
7     insertionSort(a, 5, 19)
8   return
9 }
```

```
PC = 4
l  = 5
h  = 19
p  = 46
i  = 10
```

pc	l	h	p	i
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	37	45	40	46	58	69	73	50	48	95	57	81	57
					low														high

```
0 quickSort(a, 5, 9) {
1   if(9-5>3) {
2     p = choosePivot(a, 5, 9)
3     i = partition(a, p, 5, 9)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 9)
6   } else
7     insertionSort(a, 5, 9)
8   return
9 }
```

```
PC = 0
l  = 5
h  = 9
p  = #
i  = #
```

pc	l	h	p	i
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	37	45	40	46	58	69	73	50	48	95	57	81	57
					low				high										

```
0 quickSort(a, 5, 9) {
1   if(9-5>3) {
2     p = choosePivot(a, 5, 9)
3     i = partition(a, p, 5, 9)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 9)
6   } else
7     insertionSort(a, 5, 9)
8   return
9 }
```

```
PC = 1
l  = 5
h  = 9
p  = #
i  = #
```

pc	l	h	p	i
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	37	45	40	46	58	69	73	50	48	95	57	81	57
					low				high										

```
0 quickSort(a, 5, 9) {
1   if(9-5>3) {
2     p = choosePivot(a, 5, 9)
3     i = partition(a, p, 5, 9)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 9)
6   } else
7     insertionSort(a, 5, 9)
8   return
9 }
```

```
PC = 2
l  = 5
h  = 9
p  = #
i  = #
```

pc	l	h	p	i
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	37	45	40	46	58	69	73	50	48	95	57	81	57
					low				high										

```
0 quickSort(a, 5, 9) {
1   if(9-5>3) {
2     p = choosePivot(a, 5, 9)
3     i = partition(a, p, 5, 9)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 9)
6   } else
7     insertionSort(a, 5, 9)
8   return
9 }
```

```
PC = 2
l  = 5
h  = 9
p  = #
i  = #
```

pc	l	h	p	i
2	5	9	#	#
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	**37**	45	40	46	58	69	73	50	48	95	57	81	57
					low				high										

```
0 quickSort(a, 5, 9) {
1   if(9-5>3) {
2     p = choosePivot(a, 5, 9)
3     i = partition(a, 37, 5, 9)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 9)
6   } else
7     insertionSort(a, 5, 9)
8   return
9 }
```

```
PC = 3
l  = 5
h  = 9
p  = 37
i  = #
```

pc	l	h	p	i
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	41	37	45	40	46	58	69	73	50	48	95	57	81	57
					low				high										

```
0 quickSort(a, 5, 9) {
1   if(9-5>3) {
2     p = choosePivot(a, 5, 9)
3     i = partition(a, 37, 5, 9)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 9)
6   } else
7     insertionSort(a, 5, 9)
8   return
9 }
```

```
PC = 3
l  = 5
h  = 9
p  = 37
i  = #
```

pc	l	h	p	i
3	5	9	37	#
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	**37**	41	45	40	46	58	69	73	50	48	95	57	81	57
					low				high										

```
0 quickSort(a, 5, 9) {
1   if(9-5>3) {
2     p = choosePivot(a, 5, 9)
3     i = partition(a, 37, 5, 9)
4     quickSort(a, 5, 6-1)
5     quickSort(a, 6+1, 9)
6   } else
7     insertionSort(a, 5, 9)
8   return
9 }
```

```
PC = 4
l  = 5
h  = 9
p  = 37
i  = 6
```

pc	l	h	p	i
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	41	45	40	46	58	69	73	50	48	95	57	81	57
					low				high										

```
quickSort(a, 5, 5) {
  if(5-5>3) {
    p = choosePivot(a, 5, 5)
    i = partition(a, p, 5, 5)
    quickSort(a, 5, i-1)
    quickSort(a, i+1, 5)
  } else
    insertionSort(a, 5, 5)
  return
}
```

```
PC = 0
l  = 5
h  = 5
p  = #
i  = #
```

pc	l	h	p	i
4	5	9	37	6
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	41	45	40	46	58	69	73	50	48	95	57	81	57

high

```
0 quickSort(a, 5, 5) {
1   if(5-5>3) {
2     p = choosePivot(a, 5, 5)
3     i = partition(a, p, 5, 5)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 5)
6   } else
7     insertionSort(a, 5, 5)
8   return
9 }
```

```
PC = 1
l  = 5
h  = 5
p  = #
i  = #
```

pc	l	h	p	i
4	5	9	37	6
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	41	45	40	46	58	69	73	50	48	95	57	81	57

high

```
0 quickSort(a, 5, 5) {
1   if(5-5>3) {
2     p = choosePivot(a, 5, 5)
3     i = partition(a, p, 5, 5)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 5)
6   } else
7     insertionSort(a, 5, 5)
8   return
9 }
```

```
PC = 7
l  = 5
h  = 5
p  = #
i  = #
```

pc	l	h	p	i
4	5	9	37	6
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	41	45	40	46	58	69	73	50	48	95	57	81	57

high

```
0 quickSort(a, 5, 5) {
1   if(5-5>3) {
2     p = choosePivot(a, 5, 5)
3     i = partition(a, p, 5, 5)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 5)
6   } else
7     insertionSort(a, 5, 5)
8   return
9 }
```

```
PC = 7
l  = 5
h  = 5
p  = #
i  = #
```

pc	l	h	p	i
7	5	5	#	#
4	5	9	37	6
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	41	45	40	46	58	69	73	50	48	95	57	81	57

high

```
0 quickSort(a, 5, 5) {
1   if(5-5>3) {
2     p = choosePivot(a, 5, 5)
3     i = partition(a, p, 5, 5)
4     quickSort(a, 5, i-1)
5     quickSort(a, i+1, 5)
6   } else
7     insertionSort(a, 5, 5)
8   return
9 }
```

```
PC = 8
l  = 5
h  = 5
p  = #
i  = #
```

pc	l	h	p	i
4	5	9	37	6
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	41	45	40	46	58	69	73	50	48	95	57	81	57

high

```
0 quickSort(a, 5, 9) {
1   if(9-5>3) {
2     p = choosePivot(a, 5, 9)
3     i = partition(a, 37, 5, 9)
4     quickSort(a, 5, 6-1)
5     quickSort(a, 6+1, 9)
6   } else
7     insertionSort(a, 5, 9)
8   return
9 }
```

```
PC = 5
l  = 5
h  = 9
p  = 37
i  = 6
```

pc	l	h	p	i
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	41	45	40	46	58	69	73	50	48	95	57	81	57
					low				high										

```
0 quickSort(a, 7, 9) {
1   if(9-7>3) {
2     p = choosePivot(a, 7, 9)
3     i = partition(a, p, 7, 9)
4     quickSort(a, 7, i-1)
5     quickSort(a, i+1, 9)
6   } else
7     insertionSort(a, 7, 9)
8   return
9 }
```

```
PC = 0
l  = 7
h  = 9
p  = #
i  = #
```

pc	l	h	p	i
5	5	9	37	6
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	41	45	40	46	58	69	73	50	48	95	57	81	57
							low		high										

```
0 quickSort(a, 7, 9) {
1   if(9-7>3) {
2     p = choosePivot(a, 7, 9)
3     i = partition(a, p, 7, 9)
4     quickSort(a, 7, i-1)
5     quickSort(a, i+1, 9)
6   } else
7     insertionSort(a, 7, 9)
8   return
9 }
```

```
PC = 1
l  = 7
h  = 9
p  = #
i  = #
```

pc	l	h	p	i
5	5	9	37	6
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	41	45	40	46	58	69	73	50	48	95	57	81	57
							low		high										

```
0 quickSort(a, 7, 9) {
1   if(9-7>3) {
2     p = choosePivot(a, 7, 9)
3     i = partition(a, p, 7, 9)
4     quickSort(a, 7, i-1)
5     quickSort(a, i+1, 9)
6   } else
7     insertionSort(a, 7, 9)
8   return
9 }
```

```
PC = 7
l  = 7
h  = 9
p  = #
i  = #
```

pc	l	h	p	i
5	5	9	37	6
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	41	45	40	46	58	69	73	50	48	95	57	81	57
							low		high										

```
0 quickSort(a, 7, 9) {
1   if(9-7>3) {
2     p = choosePivot(a, 7, 9)
3     i = partition(a, p, 7, 9)
4     quickSort(a, 7, i-1)
5     quickSort(a, i+1, 9)
6   } else
7     insertionSort(a, 7, 9)
8   return
9 }
```

```
PC = 7
l  = 7
h  = 9
p  = #
i  = #
```

pc	l	h	p	i
7	7	9	#	#
5	5	9	37	6
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	58	69	73	50	48	95	57	81	57
							low		high										

```
0 quickSort(a, 7, 9) {
1   if(9-7>3) {
2     p = choosePivot(a, 7, 9)
3     i = partition(a, p, 7, 9)
4     quickSort(a, 7, i-1)
5     quickSort(a, i+1, 9)
6   } else
7     insertionSort(a, 7, 9)
8   return
9 }
```

```
PC = 8
l  = 7
h  = 9
p  = #
i  = #
```

pc	l	h	p	i
5	5	9	37	6
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	58	69	73	50	48	95	57	81	57
							low		high										

```
0 quickSort(a, 5, 9) {
1   if(9-5>3) {
2     p = choosePivot(a, 5, 9)
3     i = partition(a, 37, 5, 9)
4     quickSort(a, 5, 6-1)
5     quickSort(a, 6+1, 9)
6   } else
7     insertionSort(a, 5, 9)
8   return
9 }
```

```
PC = 8
l = 5
h = 9
p = 37
i = 6
```

pc	l	h	p	i
4	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	58	69	73	50	48	95	57	81	57
					low				high										

```
0 quickSort(a, 5, 19) {
1   if(19-5>3) {
2     p = choosePivot(a, 5, 19)
3     i = partition(a, 46, 5, 19)
4     quickSort(a, 5, 10-1)
5     quickSort(a, 10+1, 19)
6   } else
7     insertionSort(a, 5, 19)
8   return
9 }
```

```
PC = 5
l  = 5
h  = 19
p  = 46
i  = 10
```

pc	l	h	p	i
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	58	69	73	50	48	95	57	81	57
					low														high

```
0 quickSort(a, 11, 19) {
1   if(19-11>3) {
2     p = choosePivot(a, 11, 19)
3     i = partition(a, p, 11, 19)
4     quickSort(a, 11, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 11, 19)
8   return
9 }
```

```
PC = 0
l  = 11
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	58	69	73	50	48	95	57	81	57
											low								high

```
0 quickSort(a, 11, 19) {
1   if(19-11>3) {
2     p = choosePivot(a, 11, 19)
3     i = partition(a, p, 11, 19)
4     quickSort(a, 11, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 11, 19)
8   return
9 }
```

```
PC = 1
l  = 11
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	58	69	73	50	48	95	57	81	57
											low								high

```
0 quickSort(a, 11, 19) {
1   if(19-11>3) {
2     p = choosePivot(a, 11, 19)
3     i = partition(a, p, 11, 19)
4     quickSort(a, 11, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 11, 19)
8   return
9 }
```

```
PC = 2
l  = 11
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	58	69	73	50	48	95	57	81	57
											low								high

```
0 quickSort(a, 11, 19) {
1   if(19-11>3) {
2     p = choosePivot(a, 11, 19)
3     i = partition(a, p, 11, 19)
4     quickSort(a, 11, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 11, 19)
8   return
9 }
```

```
PC = 2
l  = 11
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
2	11	19	#	#
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	58	69	73	50	48	95	57	81	57
											low								high

```
0 quickSort(a, 11, 19) {
1   if(19-11>3) {
2     p = choosePivot(a, 11, 19)
3     i = partition(a, 57, 11, 19)
4     quickSort(a, 11, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 11, 19)
8   return
9 }
```

```
PC = 3
l  = 11
h  = 19
p  = 57
i  = #
```

pc	l	h	p	i
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	58	69	73	50	48	95	57	81	57
											low								high

```
0 quickSort(a, 11, 19) {
1   if(19-11>3) {
2     p = choosePivot(a, 11, 19)
3     i = partition(a, 57, 11, 19)
4     quickSort(a, 11, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 11, 19)
8   return
9 }
```

```
PC = 3
l  = 11
h  = 19
p  = 57
i  = #
```

pc	l	h	p	i
3	11	19	57	#
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
											low								high

```
0 quickSort(a, 11, 19) {
1   if(19-11>3) {
2     p = choosePivot(a, 11, 19)
3     i = partition(a, 57, 11, 19)
4     quickSort(a, 11, 13-1)
5     quickSort(a, 13+1, 19)
6   } else
7     insertionSort(a, 11, 19)
8   return
9 }
```

```
PC = 4
l  = 11
h  = 19
p  = 57
i  = 13
```

pc	l	h	p	i
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
											low								high

```
0 quickSort(a, 11, 12) {
1   if(12-11>3) {
2     p = choosePivot(a, 11, 12)
3     i = partition(a, p, 11, 12)
4     quickSort(a, 11, i-1)
5     quickSort(a, i+1, 12)
6   } else
7     insertionSort(a, 11, 12)
8   return
9 }
```

```
PC = 0
l  = 11
h  = 12
p  = #
i  = #
```

pc	l	h	p	i
4	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
											low	high							

```
0 quickSort(a, 11, 12) {
1   if(12-11>3) {
2     p = choosePivot(a, 11, 12)
3     i = partition(a, p, 11, 12)
4     quickSort(a, 11, i-1)
5     quickSort(a, i+1, 12)
6   } else
7     insertionSort(a, 11, 12)
8   return
9 }
```

```
PC = 1
l  = 11
h  = 12
p  = #
i  = #
```

pc	l	h	p	i
4	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
											low	high							

```
0 quickSort(a, 11, 12) {
1   if(12-11>3) {
2     p = choosePivot(a, 11, 12)
3     i = partition(a, p, 11, 12)
4     quickSort(a, 11, i-1)
5     quickSort(a, i+1, 12)
6   } else
7     insertionSort(a, 11, 12)
8   return
9 }
```

```
PC = 7
l  = 11
h  = 12
p  = #
i  = #
```

pc	l	h	p	i
4	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
											low	high							

```
0 quickSort(a, 11, 12) {
1   if(12-11>3) {
2     p = choosePivot(a, 11, 12)
3     i = partition(a, p, 11, 12)
4     quickSort(a, 11, i-1)
5     quickSort(a, i+1, 12)
6   } else
7     insertionSort(a, 11, 12)
8   return
9 }
```

```
PC = 7
l  = 11
h  = 12
p  = #
i  = #
```

pc	l	h	p	i
7	11	12	#	#
4	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
											low	high							

```
0 quickSort(a, 11, 12) {
1   if(12-11>3) {
2     p = choosePivot(a, 11, 12)
3     i = partition(a, p, 11, 12)
4     quickSort(a, 11, i-1)
5     quickSort(a, i+1, 12)
6   } else
7     insertionSort(a, 11, 12)
8   return
9 }
```

```
PC = 8
l  = 11
h  = 12
p  = #
i  = #
```

pc	l	h	p	i
4	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
											low	high							

```
0 quickSort(a, 11, 19) {
1   if(19-11>3) {
2     p = choosePivot(a, 11, 19)
3     i = partition(a, 57, 11, 19)
4     quickSort(a, 11, 13-1)
5     quickSort(a, 13+1, 19)
6   } else
7     insertionSort(a, 11, 19)
8   return
9 }
```

```
PC = 5
l  = 11
h  = 19
p  = 57
i  = 13
```

pc	l	h	p	i
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
											low								high

```
0 quickSort(a, 14, 19) {
1   if(19-14>3) {
2     p = choosePivot(a, 14, 19)
3     i = partition(a, p, 14, 19)
4     quickSort(a, 14, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 14, 19)
8   return
9 }
```

```
PC = 0
l  = 14
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
														low					high

```
0 quickSort(a, 14, 19) {
1   if(19-14>3) {
2     p = choosePivot(a, 14, 19)
3     i = partition(a, p, 14, 19)
4     quickSort(a, 14, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 14, 19)
8   return
9 }
```

```
PC = 1
l  = 14
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
														low					high

```
0 quickSort(a, 14, 19) {
1   if(19-14>3) {
2     p = choosePivot(a, 14, 19)
3     i = partition(a, p, 14, 19)
4     quickSort(a, 14, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 14, 19)
8   return
9 }
```

```
PC = 2
l  = 14
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
														low					high

```
0 quickSort(a, 14, 19) {
1   if(19-14>3) {
2     p = choosePivot(a, 14, 19)
3     i = partition(a, p, 14, 19)
4     quickSort(a, 14, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 14, 19)
8   return
9 }
```

```
PC = 2
l  = 14
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
2	14	19	#	#
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
														low					high

```
0 quickSort(a, 14, 19) {
1   if(19-14>3) {
2     p = choosePivot(a, 14, 19)
3     i = partition(a, 69, 14, 19)
4     quickSort(a, 14, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 14, 19)
8   return
9 }
```

```
PC = 3
l  = 14
h  = 19
p  = 69
i  = #
```

pc	l	h	p	i
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	69	58	95	73	81	57
														low					high

```
0 quickSort(a, 14, 19) {
1   if(19-14>3) {
2     p = choosePivot(a, 14, 19)
3     i = partition(a, 69, 14, 19)
4     quickSort(a, 14, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 14, 19)
8   return
9 }
```

```
PC = 3
l  = 14
h  = 19
p  = 69
i  = #
```

pc	l	h	p	i
3	14	19	69	#
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
														low					high

```
0 quickSort(a, 14, 19) {
1   if(19-14>3) {
2     p = choosePivot(a, 14, 19)
3     i = partition(a, 69, 14, 19)
4     quickSort(a, 14, 16-1)
5     quickSort(a, 16+1, 19)
6   } else
7     insertionSort(a, 14, 19)
8   return
9 }
```

```
PC = 4
l  = 14
h  = 19
p  = 69
i  = 16
```

pc	l	h	p	i
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
														low					high

```
0 quickSort(a, 14, 15) {
1   if(15-14>3) {
2     p = choosePivot(a, 14, 15)
3     i = partition(a, p, 14, 15)
4     quickSort(a, 14, i-1)
5     quickSort(a, i+1, 15)
6   } else
7     insertionSort(a, 14, 15)
8   return
9 }
```

```
PC = 0
l  = 14
h  = 15
p  = #
i  = #
```

pc	l	h	p	i
4	14	19	69	16
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
														low	high				

```
0 quickSort(a, 14, 15) {
1   if(15-14>3) {
2     p = choosePivot(a, 14, 15)
3     i = partition(a, p, 14, 15)
4     quickSort(a, 14, i-1)
5     quickSort(a, i+1, 15)
6   } else
7     insertionSort(a, 14, 15)
8   return
9 }
```

```
PC = 1
l  = 14
h  = 15
p  = #
i  = #
```

pc	l	h	p	i
4	14	19	69	16
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
														low	high				

```
0 quickSort(a, 14, 15) {
1   if(15-14>3) {
2     p = choosePivot(a, 14, 15)
3     i = partition(a, p, 14, 15)
4     quickSort(a, 14, i-1)
5     quickSort(a, i+1, 15)
6   } else
7     insertionSort(a, 14, 15)
8   return
9 }
```

```
PC = 7
l  = 14
h  = 15
p  = #
i  = #
```

pc	l	h	p	i
4	14	19	69	16
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
														low	high				

```
0 quickSort(a, 14, 15) {
1   if(15-14>3) {
2     p = choosePivot(a, 14, 15)
3     i = partition(a, p, 14, 15)
4     quickSort(a, 14, i-1)
5     quickSort(a, i+1, 15)
6   } else
7     insertionSort(a, 14, 15)
8   return
9 }
```

```
PC = 7
l  = 14
h  = 15
p  = #
i  = #
```

pc	l	h	p	i
7	14	15	#	#
4	14	19	69	16
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
														low	high				

```
0 quickSort(a, 14, 15) {
1   if(15-14>3) {
2     p = choosePivot(a, 14, 15)
3     i = partition(a, p, 14, 15)
4     quickSort(a, 14, i-1)
5     quickSort(a, i+1, 15)
6   } else
7     insertionSort(a, 14, 15)
8   return
9 }
```

```
PC = 8
l  = 14
h  = 15
p  = #
i  = #
```

pc	l	h	p	i
4	14	19	69	16
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
														low	high				

```
0 quickSort(a, 14, 19) {
1   if(19-14>3) {
2     p = choosePivot(a, 14, 19)
3     i = partition(a, 69, 14, 19)
4     quickSort(a, 14, 16-1)
5     quickSort(a, 16+1, 19)
6   } else
7     insertionSort(a, 14, 19)
8   return
9 }
```

```
PC = 5
l  = 14
h  = 19
p  = 69
i  = 16
```

pc	l	h	p	i
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
														low					high

```
0 quickSort(a, 17, 19) {
1   if(19-17>3) {
2     p = choosePivot(a, 17, 19)
3     i = partition(a, p, 17, 19)
4     quickSort(a, 17, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 17, 19)
8   return
9 }
```

```
PC = 0
l  = 17
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	14	19	69	16
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
																	low		high

```
0 quickSort(a, 17, 19) {
1   if(19-17>3) {
2     p = choosePivot(a, 17, 19)
3     i = partition(a, p, 17, 19)
4     quickSort(a, 17, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 17, 19)
8   return
9 }
```

```
PC = 1
l  = 17
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	14	19	69	16
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
																	low		high

```
0 quickSort(a, 17, 19) {
1   if(19-17>3) {
2     p = choosePivot(a, 17, 19)
3     i = partition(a, p, 17, 19)
4     quickSort(a, 17, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 17, 19)
8   return
9 }
```

```
PC = 7
l  = 17
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	14	19	69	16
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
																	low		high

```
0 quickSort(a, 17, 19) {
1   if(19-17>3) {
2     p = choosePivot(a, 17, 19)
3     i = partition(a, p, 17, 19)
4     quickSort(a, 17, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 17, 19)
8   return
9 }
```

```
PC = 7
l  = 17
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
7	17	19	#	#
5	14	19	69	16
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
																	low		high

```
0 quickSort(a, 17, 19) {
1   if(19-17>3) {
2     p = choosePivot(a, 17, 19)
3     i = partition(a, p, 17, 19)
4     quickSort(a, 17, i-1)
5     quickSort(a, i+1, 19)
6   } else
7     insertionSort(a, 17, 19)
8   return
9 }
```

```
PC = 8
l  = 17
h  = 19
p  = #
i  = #
```

pc	l	h	p	i
5	14	19	69	16
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
																	low		high

```
0 quickSort(a, 14, 19) {
1   if(19-14>3) {
2     p = choosePivot(a, 14, 19)
3     i = partition(a, 69, 14, 19)
4     quickSort(a, 14, 16-1)
5     quickSort(a, 16+1, 19)
6   } else
7     insertionSort(a, 14, 19)
8   return
9 }
```

```
PC = 8
l  = 14
h  = 19
p  = 69
i  = 16
```

pc	l	h	p	i
5	11	19	57	13
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
														low					high

```
0 quickSort(a, 11, 19) {
1   if(19-11>3) {
2     p = choosePivot(a, 11, 19)
3     i = partition(a, 57, 11, 19)
4     quickSort(a, 11, 13-1)
5     quickSort(a, 13+1, 19)
6   } else
7     insertionSort(a, 11, 19)
8   return
9 }
```

```
PC = 8
l  = 11
h  = 19
p  = 57
i  = 13
```

pc	l	h	p	i
5	5	19	46	10
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
											low								high

```
0 quickSort(a, 5, 19) {
1   if(19-5>3) {
2     p = choosePivot(a, 5, 19)
3     i = partition(a, 46, 5, 19)
4     quickSort(a, 5, 10-1)
5     quickSort(a, 10+1, 19)
6   } else
7     insertionSort(a, 5, 19)
8   return
9 }
```

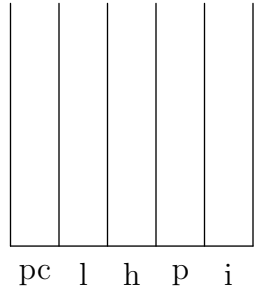
```
PC = 8
l  = 5
h  = 19
p  = 46
i  = 10
```

pc	l	h	p	i
5	0	19	28	4

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95
					low														high

```
0 quickSort(a, 0, 19) {
1   if(19-0>3) {
2     p = choosePivot(a, 0, 19)
3     i = partition(a, 28, 0, 19)
4     quickSort(a, 0, 4-1)
5     quickSort(a, 4+1, 19)
6   } else
7     insertionSort(a, 0, 19)
8   return
9 }
```

```
PC = 8
l  = 0
h  = 19
p  = 28
i  = 4
```



0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
3	18	23	24	28	32	37	40	41	45	46	48	50	57	57	58	69	73	81	95

low

high